(iii) Generation of three-phase voltages A three-phase system utilizes three separate but identical windings that are displaced by 120 electrical degrees from each other. When these three windings are rotated in an anticlockwise direction with constant angular velocity in a uniform magnetic field, the voltages are induced in each winding which have the same magnitude and frequency but are displaced 120 electrical degrees from one another.

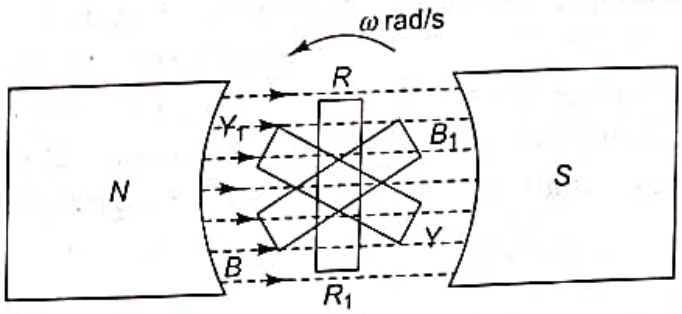


Fig. 5.7 *Three-phase system*

The instantaneous values of induced voltages in three windings RR_1, YY_1 and BB_1 are given by

$$v_R = V_m \sin \theta$$
$$v_Y = V_m \sin (\theta - 120°)$$
$$v_B = V_m \sin (\theta - 240°)$$

The induced voltage in winding YY_1 lags behind that in winding RR_1 by 120° and the induced voltage in winding BB_1 lags behind that in winding RR_1 by 240°. The waveforms of these three voltages are shown in Fig. 5.8.

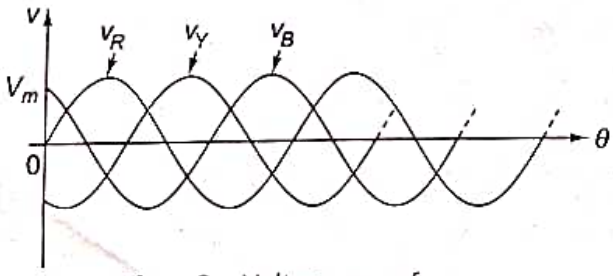


Fig. 5.8 *Voltage waveforms*

Figure 5.9 shows the phasor diagram of these induced voltages.

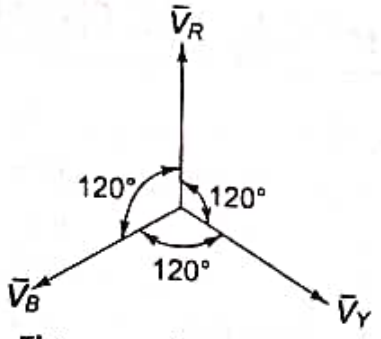


Fig. 5.9 *Phasor diagram*

5.3 ADVANTAGES OF A THREE-PHASE SYSTEM

1. In a single-phase system, the instantaneous power is fluctuating in nature. However, in a three-phase system, it is constant at all times.
2. The output of a three-phase system is greater than that of a single-phase system.
3. Transmission and distribution of a three-phase system is cheaper than that of a single-phase system.
4. Three-phase motors are more efficient and have higher power factors than single-phase motors of the same frequency.
5. Three-phase motors are self-starting whereas single-phase motors are not self-starting.

5.4 SOME DEFINITIONS

Phase Sequence The sequence in which the voltages in the three phases reach the maximum positive value is called the *phase sequence* or *phase order*. From the phasor diagram of a three-phase system, it is clear that the voltage in the coil *R* attains maximum positive value first, next in the coil *Y* and then in the coil *B*. Hence, the phase sequence is *R-Y-B*.

Phase Voltage The voltage induced in each winding is called the *phase voltage*.

Phase Current The current flowing through each winding is called the *phase current*.

Line Voltage The voltage available between any pair of terminals or lines is called the *line voltage*.

Line Current The current flowing through each line is called the *line current*.

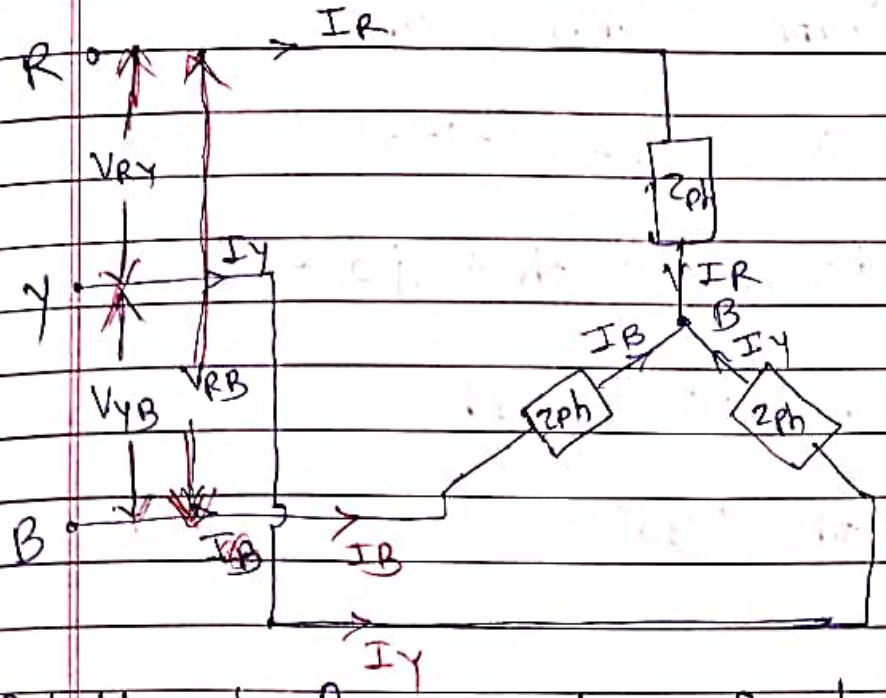
Symmetrical or Balanced System A three-phase system is said to be balanced if the

(a) voltages in the three phases are equal in magnitude and differ in phase from one another by 120°, and

(b) currents in the three phases are equal in magnitude and differ in phase from one another by 120°.

Balanced Load The load is said to be balanced if loads connected across the three phases are identical, i.e., all the loads have the same magnitude and power factor.

(*) Voltage, current & power Relation in a Balanced star-connected load.



→ Relation betn line voltage & phase voltage (in ↲)

> since the system is balanced, the 3-ϕ voltage V_{RN}, V_{YN}, & V_{BN} are equal in magnitude & differ in phase from one anothe by 120°

$V_{RN} = V_{YN} = V_{BN} = V_{ph}$ ---- phase vtg

where as

V_{ph} indicate the rms value of phase voltage

$V_{RN} = V_{ph} \angle 0°$

$V_{YN} = V_{ph} \angle -120°$

$V_{BN} = V_{ph} \angle -240°$

$V_{RY} = V_{YB} = V_{RB} = V_L$ ----- line vtg

let

where V_L indicate line voltage

Apply kirchhoff's ~~law~~ voltage law (RY)

$$\overline{V_{RY}} = \overline{V_{RN}} + \overline{V_{YN}} = \overline{V_{RN}} - \overline{V_{YN}}$$

$V_{RY} = V_R - V_Y$

$= V_R + (-V_Y) = V_{ph} \angle 0^\circ - V_{ph} \angle -120^\circ$

$= V_{ph} + j0 + 0.5 V_{ph} + j0.866 V_{ph}$

$V_{YB} = V_Y - V_B$

$= V_Y + (-V_B)$

$= 1.5 V_{ph} + j0.866 V_{ph}$

$1.73 V_{ph} \angle 30$

$= \sqrt{3} V_{ph} \angle 30^\circ$

Similarly

$$\overline{V_{YB}} = \overline{V_{YN}} + \overline{V_{BN}} = \sqrt{3} V_{ph} \angle 30^\circ$$

$$V_{BR} = \overline{V_{BN}} + \overline{V_{RN}} = \sqrt{3} V_{ph} \angle 30^\circ$$

Thus in a star connected, 3-ϕ system $\boxed{V_L = \sqrt{3} V_{ph}}$

& line voltage lead respective phase voltages by 30°

→ Relation betn line current & phase current (in Y)

$$\boxed{I_L = I_{ph}}$$

→ Power

total power = P = 3 * power in each phase

$$P = 3 * V_{ph} * I_{ph} * \cos\phi$$

$V_{ph} =$

$V_L = \sqrt{3} V_{ph}$ & $I_L = I_{ph}$

$$V_{ph} = \frac{V_L}{\sqrt{3}}$$

$$P = 3 * \frac{V_L}{\sqrt{3}} * I_L * \cos\phi$$

$$P = \sqrt{3}\, V_L I_L \cos\phi$$

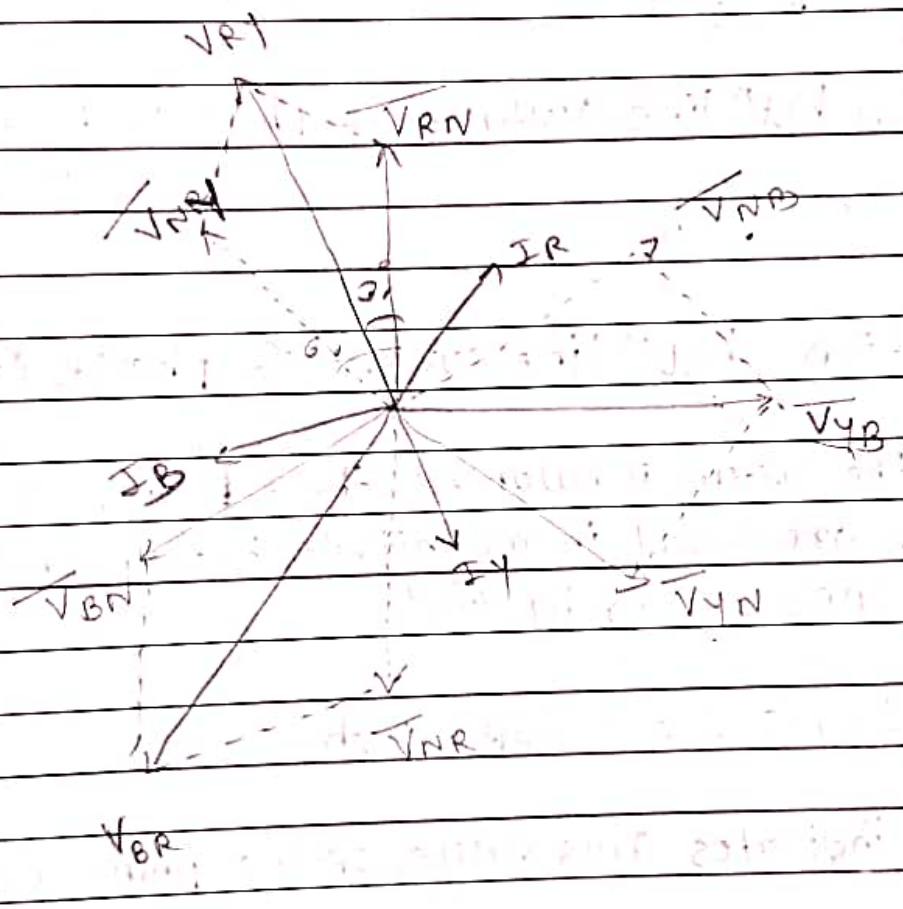
Similarly $Q = 3\, V_{ph} I_{ph} \sin\phi$

$= \sqrt{3}\, V_L I_L \sin\phi$

Similarly $S = 3\, V_{ph} I_{ph}$

$= \sqrt{3}\, V_{ph} I_{ph}$

(law of parallelogram)



Steps for Drawing Phasor Diagram

1. First draw $\overline{V}_{RN}$ as reference voltage.
2. Since three phase voltages are equal in magnitude and differ in phase from one another by 120°, draw $\overline{V}_{YN}$ and $\overline{V}_{BN}$ lagging 120° behind w.r.t. each other.
3. Draw $\overline{V}_{NY}$ equal and opposite to $\overline{V}_{YN}$.
4. Add $\overline{V}_{RN}$ and $\overline{V}_{NY}$ using the parallelogram law of vector addition such that
$$\overline{V}_{RY} = \overline{V}_{RN} + \overline{V}_{NY}$$
5. Draw $\overline{V}_{NB}$ equal and opposite to $\overline{V}_{BN}$.

6. Add $\overline{V}_{YN}$ and $\overline{V}_{NB}$ using the parallelogram law of vector addition such that
$$\overline{V}_{YB} = \overline{V}_{YN} + \overline{V}_{NB}$$
7. Draw $\overline{V}_{NR}$ equal and opposite to $\overline{V}_{RN}$.
8. Add $\overline{V}_{BN}$ and $\overline{V}_{NR}$ using the parallelogram law of vector addition such that
$$\overline{V}_{BR} = \overline{V}_{BN} + \overline{V}_{NR}$$
9. Assuming inductive load, draw three phase currents $\overline{I}_R$, $\overline{I}_Y$ and $\overline{I}_B$ lagging behind its respective phase voltages by an angle ϕ. The phase currents are equal in magnitude and differ in phase from one another by 120°.
10. Line currents are same as the phase currents in star connected load. Hence, separate line currents are not drawn.
11. Since $\overline{V}_{NY}$ is antiphase with $\overline{V}_{YN}$, angle between $\overline{V}_{RN}$ and $\overline{V}_{NY}$ is 60°. The line voltage $\overline{V}_{RY}$ leads phase voltage $\overline{V}_{RN}$ by 30°. Similarly, line voltage $\overline{V}_{YB}$ leads phase voltage $\overline{V}_{YN}$ by 30° and line voltage $\overline{V}_{BR}$ leads phase voltage $\overline{V}_{BN}$ by 30°.

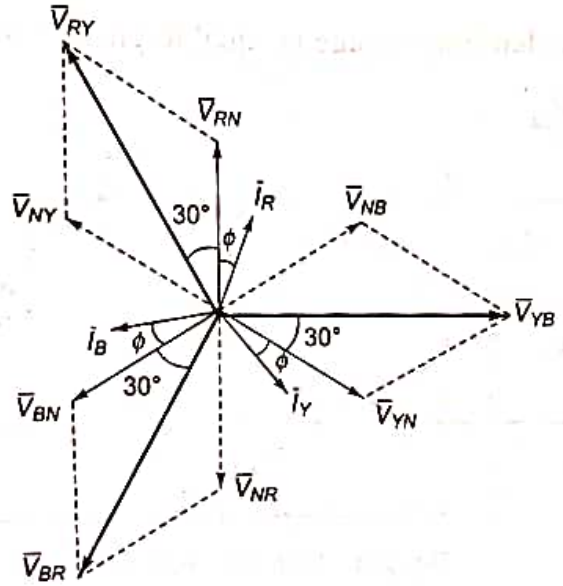


Fig. 5.16 *Phasor diagram*

5.8.4 Power

The total power in a three-phase system is the sum of powers in the three phases. For a balanced load, the power consumed in each load phase is the same.

Total active power $P = 3 \times$ power in each phase $= 3\, V_{ph}\, I_{ph} \cos\phi$

In a star-connected, three-phase system,

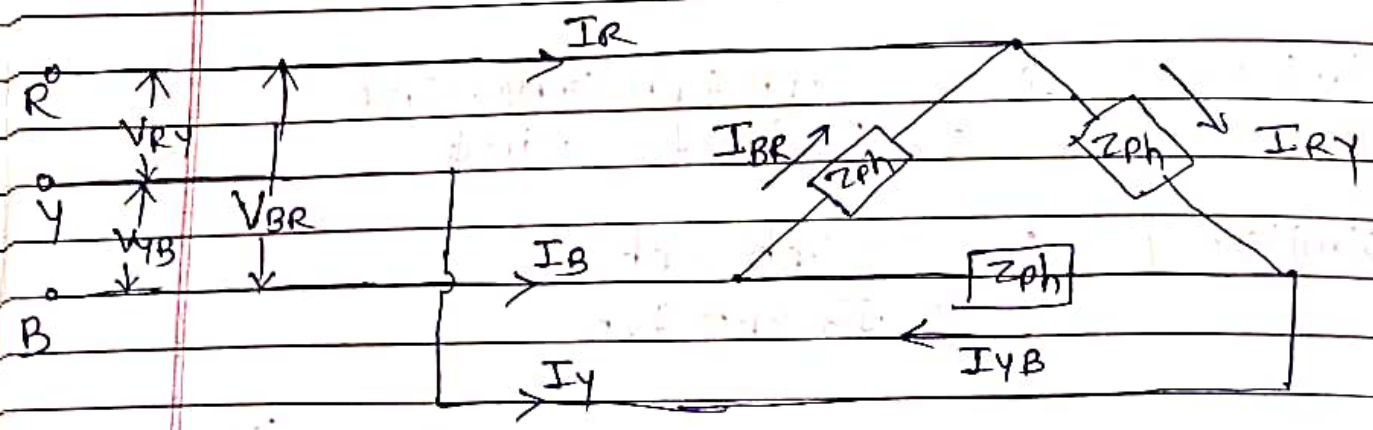
$$V_{ph} = \frac{V_L}{\sqrt{3}}$$

$$I_{ph} = I_L$$

$$P = 3 \times \frac{V_L}{\sqrt{3}} \times I_L \times \cos\phi = \sqrt{3}\, V_L\, I_L \cos\phi$$

where ϕ is the phase difference between phase voltage and corresponding phase current.

(*) Voltage, current & power Relation in Balanced Delta connected load



→ Relation betⁿ line voltage & phase voltage

$$\boxed{V_L = V_{ph}}$$

→ Relation betⁿ line current & phase current

Since the system is balanced, the 3-ϕ phase current I_{RY}, I_{YB} & I_{BR} are equal in magnitude but differ in phase from one another by 120°

Let

$$I_{RY} = I_{YB} = I_{BR} = I_{ph}$$

where

I_{ph} indicates rms value of the phase current

$$I_{RY} = I_{ph} \angle 0$$

$$I_{YB} = I_{ph} \angle -120^\circ$$

$$I_{BR} = I_{ph} \angle -240$$

$$I_R = I_Y = I_B = I_L$$

where

I_L indicate rms value of the line current

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apply kirchhoff's current law

$$\overline{I_R} + \overline{I_{BR}} = \overline{I_{RY}}$$

$$\overline{I_R} = \overline{I_{RY}} - \overline{I_{BR}}$$

$$= I_{ph} \angle 0 - I_{ph} \angle -240$$

$$= I_{ph} + j0 + 0.5 I_{ph} - j0.866 I_{ph}$$

$$= \sqrt{3}\, I_{ph} \angle -30$$

Similarly

$$\overline{I_Y} = \overline{I_{YB}} - \overline{I_{RY}} = \sqrt{3}\, I_{ph} \angle -30^\circ$$

$$I_B = I_{BR} - I_{YB} = \sqrt{3}\, I_{ph} \angle -30^\circ$$

Thus in delta connected 3-ϕ system $\boxed{I_L = \sqrt{3} I_{ph}}$

→ power

total power $P = 3\, V_{ph} I_{ph} \cos\phi$

in delta connected $V_{ph} = V_L$ & $I_L = \sqrt{3} I_{ph}$.

$$P = 3 * V_L * \frac{I_L}{\sqrt{3}} * \cos\phi = \sqrt{3}\, V_L I_L \cos\phi$$

$$Q = \sqrt{3}\, V_L I_L \cos\phi$$

$$S = \sqrt{3}\, V_L I_L$$

Steps for Drawing Phasor Diagram

1. First draw $\overline{V}_{RY}$ as reference voltage.
2. Since three phase voltages are equal in magnitude and differ in phase from one another by 120°, draw $\overline{V}_{YB}$ and $\overline{V}_{BR}$ lagging 120° behind w.r.t. each other.
3. Line voltages are same as the phase voltages for balanced delta connected load. Hence, separate line voltages are not drawn.
4. Assuming inductive load, draw three phase currents $\overline{I}_{RY}$, $\overline{I}_{YB}$ and $\overline{I}_{BR}$ lagging behind respective phase voltages by an angle ϕ. The phase currents are equal in magnitude and differ in phase from one another by 120°.
5. Draw $-\overline{I}_{BR}$ equal and opposite to $\overline{I}_{BR}$.
6. Add $\overline{I}_{RY}$ and $-\overline{I}_{BR}$ using the parallelogram law of vector addition such that
$$\overline{I}_R = \overline{I}_{RY} - \overline{I}_{BR}$$
7. Draw $-\overline{I}_{RY}$ equal and opposite to $\overline{I}_{RY}$.
8. Add $\overline{I}_{YB}$ and $-\overline{I}_{RY}$ using the parallelogram law of vector addition such that
$$\overline{I}_Y = \overline{I}_{YB} - \overline{I}_{RY}$$
9. Draw $-\overline{I}_{YB}$ equal and opposite to $\overline{I}_{YB}$.
10. Add $\overline{I}_{BR}$ and $-\overline{I}_{YB}$ using the parallelogram laws of vector addition such that
$$\overline{I}_B = \overline{I}_{BR} - \overline{I}_{YB}$$
11. Since $-\overline{I}_{BR}$ is antiphase with $\overline{I}_{BR}$, angle between $\overline{I}_{RY}$ and $-\overline{I}_{BR}$ is 60°. The line current $\overline{I}_R$ lags behind phase current $\overline{I}_{RY}$ by 30°. Similarly, the line current $\overline{I}_Y$ lags behind phase current $\overline{I}_{YB}$ by 30° and the line current $\overline{I}_B$ lags behind phase current $\overline{I}_{BR}$ by 30°.

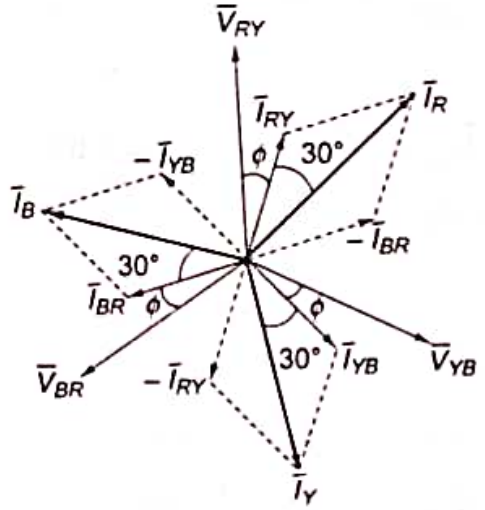


Fig. 5.19 Phasor diagram

5.9.4 Power

$$P = 3\, V_{ph}\, I_{ph} \cos \phi$$

In a delta-connected, three-phase system,

$$V_{ph} = V_L$$

$$I_{ph} = \frac{I_L}{\sqrt{3}}$$

$$P = 3 \times V_L \times \frac{I_L}{\sqrt{3}} \times \cos \phi = \sqrt{3}\ V_L\, I_L \cos \phi$$

Total reactive power $Q = 3\, V_{ph}\, I_{ph} \sin \phi = \sqrt{3}\ V_L\, I_L \sin \phi$

Total apparent power $S = 3\, V_{ph}\, I_{ph} = \sqrt{3}\ V_L\, I_L$

The power triangle for a three-phase system is shown in Fig. 5.20.

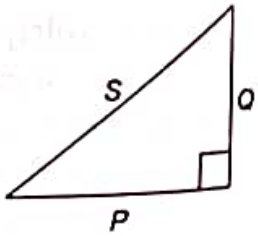
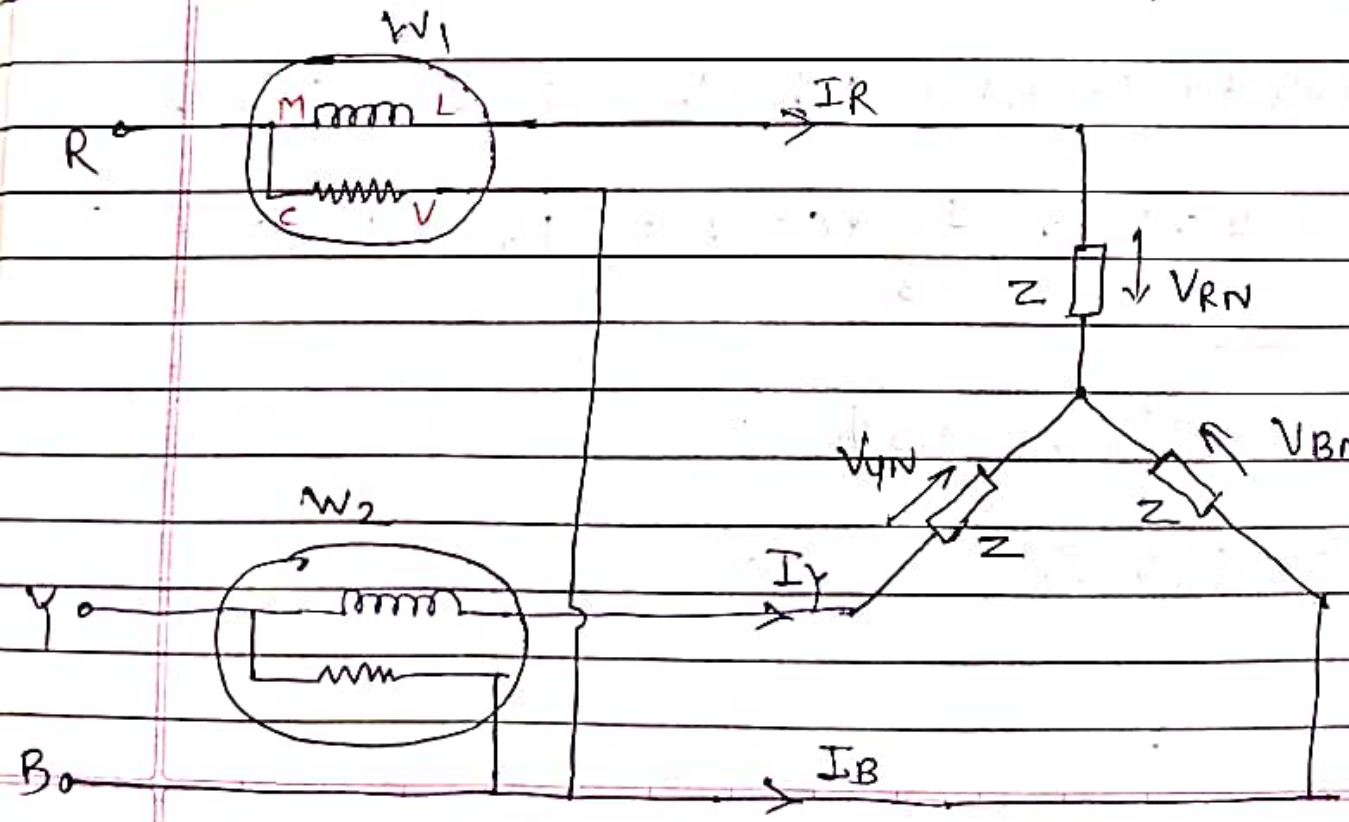


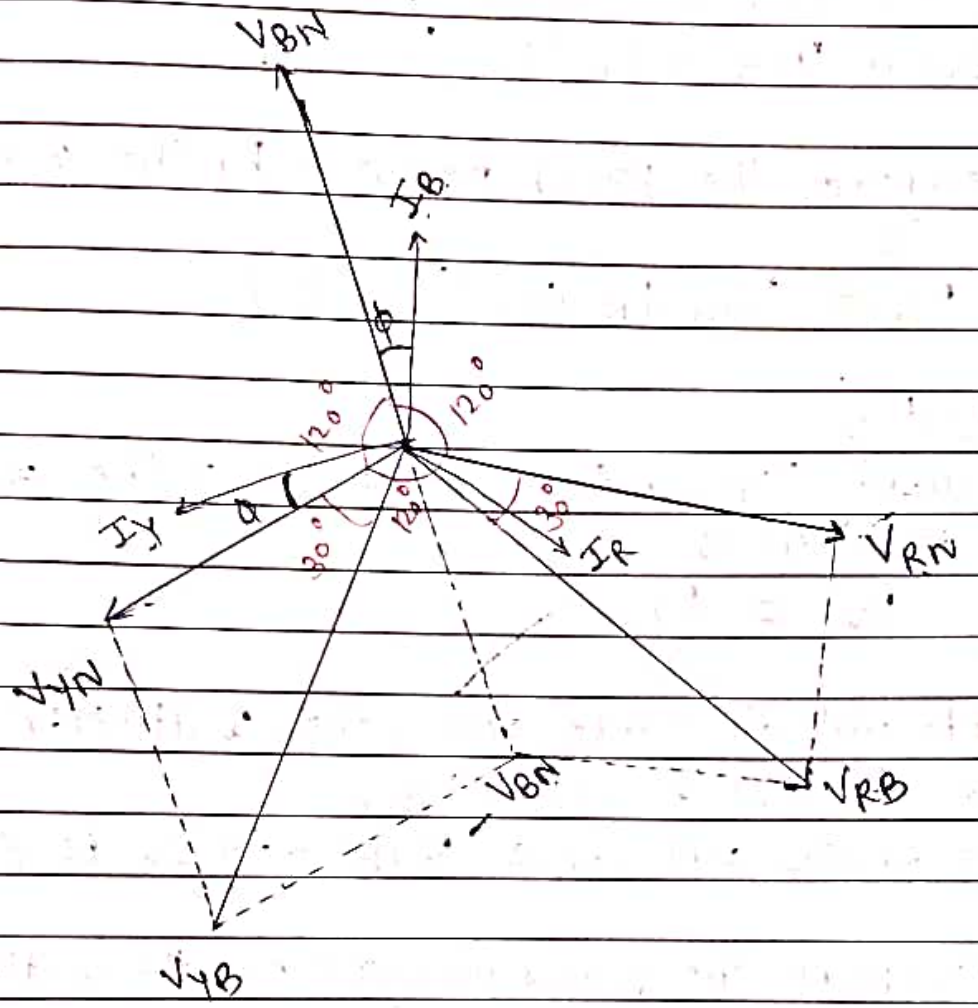
Fig. 5.20

(*) two wattmeter Methode VIMP

> The connection diagram of a 3-ϕ balanced load connected as star connection is shown below



> The load is considered as an inductive load, and thus, the phasor diagram of the inductive load is drawn below.



> The three phase voltages V_{RN}, V_{YN} & V_{BN}, are displaced by an angle of 120° electrical as shown in the phasor diagram

> The phase current lag behind their respective phase voltage by an angle ϕ

> Now the current flowing through the current coil of the Wattmeter W_1 will be given as

$$W_1 = I_R$$

> Potential difference across the pressure or potential coil of the wattmeter W_1 will be

$$W_1 = V_{RB} = V_{RN} - V_{BN}$$

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> To obtain the value of V_{YB} reverse the phasor V_{BN} and add in to the phasor V_{YN} as shown in the phasor diagram. The phase difference between V_{RB} & I_R is $(30^\circ - \phi)$

> Therefore, the power measured by the wattmeter W_1 is

$$W_1 = V_{RB} I_R \cos(30 - \phi)$$

Second wattmeter

> current through the current coil of the wattmeter W_2 is given as

$$W_2 = I_Y$$

> potential difference across the wattmeter W_2 is

$$W_2 = V_{YB} = V_{RN} - V_{BN}$$

> The phase difference V_{YB} and I_Y is $(30 + \phi)$

> Therefore, the power measured by the wattmeter

$$W_2 = V_{YB} * I_Y \cos(30 + \phi)$$

> Since the load is in balanced condition, hence

$$I_R = I_Y = I_B = I_L$$
$$V_{RY} = V_{YB} = V_{BR} = V_L$$

> Therefor wattmeter the wattmeter reading will be

$$W_1 = V_L I_L \cos(30^\circ - \phi)$$
$$W_2 = V_L I_L \cos(30^\circ + \phi)$$

> Now the sum of the two wattmeter reading will be given as

$\cos(x \pm y) = \cos x \cos y \pm \sin x \sin y$

$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$

$W_1 + W_2 = [V_L I_L \cos(30^\circ - \phi) + V_L I_L \cos(30^\circ + \phi)]$

$W_1 + W_2 = [V_L I_L [\cos 30^\circ \cdot \cos\phi + \sin 30^\circ \cdot \sin\phi]$

$\cos(x - y) = \cos x \cos y + \sin x \sin y$

$\cos(x + y) = \cos x \cos y - \sin x \sin y$

$W_1 + W_2 = V_L I_L \cos(30^\circ - \phi) + V_L I_L \cos(30 + \phi)$

$W_1 + W_2 = V_L I_L [\cos(30^\circ - \phi) + \cos(30^\circ + \phi)]$

$W_1 + W_2 = V_L I_L [\cos 30^\circ \cos\phi + \sin 30^\circ \sin\phi$

$+ \cos 30^\circ \cos\phi - \sin 30^\circ \sin\phi]$

$W_1 + W_2 = V_L I_L (2 \cos 30^\circ \cos\phi)$

$W_1 + W_2 = V_L I_L (\cancel{2} \frac{\sqrt{3}}{\cancel{2}} \cos\phi)$

$W_1 + W_2 = \sqrt{3} V_L I_L \cos\phi$ — (1)

→ As we know that

$W_1 + W_2 = \sqrt{3} V_L I_L \cos\phi$

Now

$W_1 - W_2 = [V_L I_L \cos(30^\circ - \phi) - V_L I_L \cos(30^\circ + \phi)]$

$W_1 - W_2 = V_L I_L (\cos 30^\circ \cdot \cos\phi + \sin 30^\circ \cdot \sin\phi$

$- \cos 30^\circ \cos\phi + \sin 30^\circ \sin\phi)$

$W_1 - W_2 = V_L I_L (2 \sin 30^\circ \sin\phi)$

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$W_1 - W_2 = 2 V_L I_L \sin 30 . \sin\phi$

$W_1 - W_2 = 2 V_L I_L \frac{\sqrt{3}}{2} \sin\phi$

$W_1 - W_2 = V_L I_L \sin\phi$ —— (2)

Determination of Reactive power by two wattmeter methode.

> To get the reactive power, multiply eq^n (2) $\sqrt{3}$

$\sqrt{3} (W_1 - W_2) = \sqrt{3} V_L I_L \sin\phi = P_r$

Therefor, the reactive power is given by the eq^n shown below

$$P_r = \sqrt{3} (W_1 - W_2)$$

Example 6

Three coils, each having a resistance of 8 Ω and an inductance of 0.02 H, are connected in delta to a three-phase, 400 V, 50 Hz supply. Calculate the (i) line current, and (ii) power absorbed.

Solution

$$R = 8\ \Omega$$
$$L = 0.02\ \text{H}$$
$$V_L = 400\ \text{V}$$
$$f = 50\ \text{Hz}$$

For a delta-connected load,

(i) Line current

$$V_L = V_{ph} = 400\ \text{V}$$
$$X_L = 2\pi fL = 2\pi \times 50 \times 0.02 = 6.28\ \Omega$$
$$\overline{Z}_{ph} = R + jX_L = 8 + j6.28 = 10.17\ \angle 38.13°\ \Omega$$
$$Z_{ph} = 10.17\ \Omega$$
$$\phi = 38.13°$$
$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{400}{10.17} = 651639.33\ \text{A}$$
$$I_L = \sqrt{3}\ I_{ph} = \sqrt{3} \times 39.33 = 68.12\ \text{A}$$

(ii) Power absorbed

$$P = \sqrt{3}\ V_L I_L \cos\phi = \sqrt{3} \times 400 \times 68.12 \times \cos(38.13°) = 37.12\ \text{kW}$$

Example 7

The three equal impedances of each of 10 ∠60° Ω, are connected in star across a three-phase, 400 V, 50 Hz supply. Calculate the (i) line voltage and phase voltage, (ii) power factor and active power consumed, (iii) If the same three impedances are connected in delta to the same source of supply, what is the active power consumed?

Solution

$$\bar{Z}_{ph} = 10 \angle 60° \ \Omega$$
$$V_L = 400 \text{ V}$$
$$f = 50 \text{ Hz}$$

For a star-connected load,

(i) Line voltage and phase voltage

$$V_L = 400 \text{ V}$$
$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$$

(ii) Power factor and active power consumed

$$\phi = 60°$$
$$\text{pf} = \cos\phi = \cos(60°) = 0.5 \text{ (lagging)}$$
$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{230.94}{10} = 23.094 \text{ A}$$
$$I_L = I_{ph} = 23.094 \text{ A}$$
$$P = \sqrt{3}\, V_L I_L \cos\phi = \sqrt{3} \times 400 \times 23.094 \times 0.5 = 8 \text{ kW}$$

(iii) Active power consumed for delta-connected load

$$V_L = 400 \text{ V}$$
$$Z_{ph} = 10 \ \Omega$$
$$V_{ph} = V_L = 400 \text{ V}$$
$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{400}{10} = 40 \text{ A}$$
$$I_L = \sqrt{3}\, I_{ph} = \sqrt{3} \times 40 = 69.28 \text{ A}$$
$$P = \sqrt{3}\, V_L I_L \cos\phi = \sqrt{3} \times 400 \times 69.28 \times \cos(60°) = 24 \text{ kW}$$

Example 8

Three similar coils A, B, and C are available. Each coil has a 9 Ω resistance and a 12 Ω reactance. They are connected in delta to a three-phase, 440 V, 50 Hz supply. Calculate for this load, the (i) phase current, (ii) line current, (iii) power factor, (iv) total kVA, (v) active power, and (vi) reactive power. If these coils are connected in star across the same supply, calculate all the above quantities.

Solution

$$R = 9 \ \Omega$$
$$X_L = 12 \ \Omega$$
$$V_L = 440 \text{ V}$$
$$f = 50 \text{ Hz}$$

For a delta-connected load,

(i) Phase current

$$V_L = V_{ph} = 440 \text{ V}$$
$$\bar{Z}_{ph} = 9 + j12 = 15 \angle 53.13° \ \Omega$$
$$Z_{ph} = 15 \ \Omega$$
$$\phi = 53.13°$$
$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{440}{15} = 29.33 \text{ A}$$

(ii) Line current

$$I_L = \sqrt{3} \ I_{ph} = \sqrt{3} \times 29.33 = 50.8 \text{ A}$$

(iii) Power factor

$$\text{pf} = \cos \phi = \cos (53.13°) = 0.6 \text{ (lagging)}$$

(iv) Total kVA

$$S = \sqrt{3} \ V_L I_L = \sqrt{3} \times 440 \times 50.8 = 38.71 \text{ kVA}$$

(v) Active power

$$P = \sqrt{3} \ V_L I_L \cos \phi = \sqrt{3} \times 440 \times 50.8 \times 0.6 = 23.23 \text{ kW}$$

(vi) Reactive power

$$Q = \sqrt{3} \ V_L I_L \sin \phi = \sqrt{3} \times 440 \times 50.8 \times \sin (53.13°) = 30.97 \text{ kVAR}$$

If these coils are connected in star across the same supply,

(i) Phase current

$$V_L = 440 \text{ V}$$
$$Z_{ph} = 15 \ \Omega$$
$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{440}{\sqrt{3}} = 254.03 \text{ V}$$
$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{254.03}{15} = 16.94 \text{ A}$$

(ii) Line current

$$I_L = I_{ph} = 16.94 \text{ A}$$

(iii) Power factor

$$\text{pf} = 0.6 \text{ (lagging)}$$

(iv) Total kVA

$$S = \sqrt{3} \ V_L I_L = \sqrt{3} \times 440 \times 16.94 = 12.91 \text{ kVA}$$

(v) Active power

$$P = \sqrt{3}\ V_L I_L \cos\phi = \sqrt{3} \times 440 \times 16.94 \times 0.6 = 7.74 \text{ kW}$$

(vi) Reactive power

$$Q = \sqrt{3}\ V_L I_L \sin\phi = \sqrt{3} \times 440 \times 16.94 \times \sin(53.13°) = 12.33 \text{ kVAR}$$

Example 9

A balanced 3-phase load consists of 3 coils, each of resistance 4 Ω and inductance 0.02 H. It is connected to a 440 V, 50 Hz, 3 ϕ supply. Find the total power consumed when the load is connected in star and the total reactive power when the load is connected in delta. **[Dec 2014]**

Solution

$$R = 4\ \Omega$$
$$L = 0.02 \text{ H}$$
$$V_L = 440 \text{ V}$$
$$f = 50 \text{ Hz}$$

For a star-connected load,

(i) Total power consumed

$$X_L = 2\pi f L = 2\pi \times 50 \times 0.02 = 6.28\ \Omega$$

$$\bar{Z}_{ph} = R + j X_L = 4 + j\,6.28 = 7.45 \angle 57.51°\ \Omega$$

$$Z_{ph} = 7.45\ \Omega$$

$$\phi = 57.51°$$

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{440}{\sqrt{3}} = 254.03 \text{ V}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{254.03}{7.45} = 34.1 \text{ A}$$

$$I_L = I_{ph} = 34.1 \text{ A}$$

$$P = \sqrt{3}\ V_L I_L \cos\phi = \sqrt{3} \times 440 \times 34.1 \times \cos(57.51°) = 13.96 \text{ kW}$$

$$Q = \sqrt{3}\ V_L I_L \sin\phi = \sqrt{3} \times 440 \times 34.1 \times \sin(57.51°) = 21.92 \text{ kVAR}$$

(ii) When the load is connected in delta across same supply

$$Q_\Delta = 3Q_Y = 3 \times 21.92 \times 10^3 = 65.76 \text{ kVAR}$$

Example 10

A 415 V, 50 Hz, three-phase voltage is applied to three star-connected identical impedances. Each impedance consists of a resistance of 15 Ω, a capacitance of 177 μF and an inductance of 0.1 henry in series. Find the (i) power factor, (ii) phase current, (iii) line current, (iv) active power, (v) reactive power, and (vi) total VA. Draw a neat phasor diagram. If the same impedances are connected in delta, find the (i) line current, and (ii) power consumed. **[Dec 2015]**

Solution

$$V_L = 415 \text{ V}$$
$$f = 50 \text{ Hz}$$
$$R = 15\ \Omega$$
$$C = 177\ \mu\text{F}$$
$$L = 0.1 \text{ H}$$

For a star-connected load,

(i) Power factor

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.1 = 31.42\ \Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 177 \times 10^{-6}} = 17.98\,\Omega$$

$$\overline{Z}_{ph} = R + jX_L - jX_C$$
$$= 15 + j31.42 - j17.98$$
$$= 15 + j13.44$$
$$= 20.14\ \angle 41.86°\ \Omega$$
$$Z_{ph} = 20.14\ \Omega$$
$$\phi = 41.86°$$
$$\text{pf} = \cos\phi = \cos(41.86°) = 0.744 \text{ (lagging)}$$

(ii) Phase current

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{415}{\sqrt{3}} = 239.6\,\text{V}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{239.6}{20.14} = 11.9\,\text{A}$$

(iii) Line current

$$I_L = I_{ph} = 11.9 \text{ A}$$

(iv) Active power

$$P = \sqrt{3}\ V_L I_L \cos\phi = \sqrt{3} \times 415 \times 11.9 \times 0.744 = 6.36 \text{ kW}$$

(v) Reactive power

$$Q = \sqrt{3}\ V_L I_L \sin\phi = \sqrt{3} \times 415 \times 11.9 \times \sin(41.86°) = 5.71 \text{ kVAR}$$

(vi) Total VA

$$S = \sqrt{3}\ V_L I_L = \sqrt{3} \times 415 \times 11.9 = 8.55 \text{ kVA}$$

Phasor Diagram

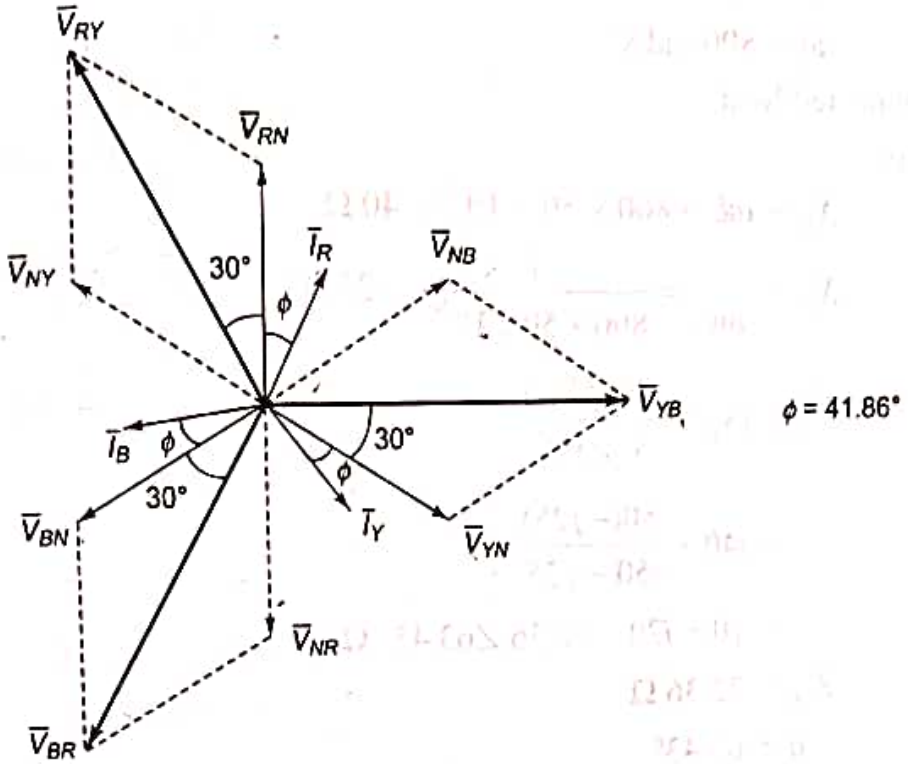

Fig. 5.21

If the same impedances are connected in delta,

(i) Line current

$$V_L = V_{ph} = 415 \text{ V}$$

$$Z_{ph} = 20.14\ \Omega$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{415}{20.14} = 20.61 \text{ A}$$

$$I_L = \sqrt{3}\ I_{ph} = \sqrt{3} \times 20.61 = 35.69 \text{ A}$$

(ii) Power consumed

$$P = \sqrt{3}\ V_L I_L \cos\phi = \sqrt{3} \times 415 \times 35.69 \times 0.744 = 19.09 \text{ kW}$$

Example 11

Each phase of a delta-connected load consists of a 50 mH inductor in series with a parallel combination of a 50 Ω resistor and a 50 μF capacitor. The load is connected to a three-phase, 550 V, 800 rad/s ac supply. Find the (i) power factor, (ii) phase current, (iii) line current, (iv) power drawn, (v) reactive power, and (vi) kVA rating of the load.

Solution

$$L = 50 \text{ mH}$$

$$R = 50\ \Omega$$

$$C = 50\ \mu\text{F}$$

$$V_L = 550 \text{ V}$$
$$\omega = 800 \text{ rad/s}$$

For a delta-connected load,

(i) Power factor

$$X_L = \omega L = 800 \times 50 \times 10^{-3} = 40\ \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{800 \times 50 \times 10^{-6}} = 25\Omega$$

$$\overline{Z}_{ph} = jX_L + \frac{R(-jX_C)}{R - jX_C}$$

$$= j40 + \frac{50(-j25)}{50 - j25}$$

$$= 10 + j20 = 22.36 \angle 63.43°\ \Omega$$

$$Z_{ph} = 22.36\ \Omega$$

$$\phi = 63.43°$$

$$\text{pf} = \cos\phi = \cos(63.43°) = 0.447 \text{ (lagging)}$$

(ii) Phase current

$$V_L = V_{ph} = 550 \text{ V}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{550}{22.36} = 24.6 \text{ A}$$

(iii) Line current

$$I_L = \sqrt{3}\ I_{ph} = \sqrt{3} \times 24.6 = 42.61 \text{ A}$$

(iv) Power drawn

$$P = \sqrt{3}\ V_L I_L \cos\phi = \sqrt{3} \times 550 \times 42.61 \times 0.447 = 18.14 \text{ kW}$$

(v) Reactive power

$$Q = \sqrt{3}\ V_L I_L \sin\phi = \sqrt{3} \times 550 \times 42.61 \times \sin(63.43°) = 36.3 \text{ kVAR}$$

(vi) kVA rating of the load

$$S = \sqrt{3}\ V_L I_L = \sqrt{3} \times 550 \times 42.61 = 40.59 \text{ kVA}$$

Example 15

Three identical coils connected in delta to a 440 V, three-phase supply take a total power of 50 kW and a line current of 90 A. Find the (i) phase current, (ii) power factor, and (iii) apparent power taken by the coils.

Solution

$$V_L = 440 \text{ V}$$
$$P = 50 \text{ kW}$$
$$I_L = 90 \text{ A}$$

For a delta-connected load,

(i) Phase current

$$I_{ph} = \frac{I_L}{\sqrt{3}} = \frac{90}{\sqrt{3}} = 51.96 \text{ A}$$

(ii) Power factor

$$P = \sqrt{3}\, V_L I_L \cos\phi$$
$$50 \times 10^3 = \sqrt{3} \times 440 \times 90 \times \cos\phi$$
$$\text{pf} = \cos\phi = 0.73 \text{ (lagging)}$$

(iii) Apparent power

$$S = \sqrt{3}\, V_L I_L = \sqrt{3} \times 440 \times 90 = 68.59 \text{ kVA}$$

Example 16

Three similar choke coils are connected in star to a three-phase supply. If the line current is 15 A, the total power consumed is 11 kW and the volt-ampere input is 15 kVA, find the line and phase voltages, the VAR input and the reactance and resistance of each coil. If these coils are now connected in delta to the same supply, calculate phase and line currents, active and reactive power.

Solution

$$I_L = 15 \text{ A}$$
$$P = 11 \text{ kW}$$
$$S = 15 \text{ kVA}$$

For a star-connected load,

(i) Line voltage

$$S = \sqrt{3}\, V_L I_L$$

$$15 \times 10^3 = \sqrt{3} \times V_L \times 15$$
$$V_L = 577.35 \text{ V}$$

(ii) Phase voltage

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{577.35}{\sqrt{3}} = 333.33 \text{ V}$$

(iii) VAR input

$$\cos\phi = \frac{P}{S} = \frac{11 \times 10^3}{15 \times 10^3} = 0.733$$
$$\phi = 42.86°$$
$$Q = \sqrt{3}\ V_L I_L \sin\phi = \sqrt{3} \times 577.35 \times 15 \times \sin(42.86°) = 10.2 \text{ kVAR}$$

(iv) Reactance and resistance of coil

$$I_{ph} = I_L = 15 \text{ A}$$
$$Z_{ph} = \frac{V_{ph}}{I_{ph}} = \frac{333.33}{15} = 22.22\ \Omega$$
$$R = Z_{ph} \cos\phi = 22.22 \times 0.733 = 16.29\ \Omega$$
$$X_L = Z_{ph} \sin\phi = 22.22 \times \sin(42.86°) = 15.11\ \Omega$$

If these coils are now connected in delta,

(i) Phase current

$$V_{ph} = V_L = 577.35 \text{ V}$$
$$Z_{ph} = 22.22\ \Omega$$
$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{577.35}{22.22} = 25.98 \text{ A}$$

(ii) Line current

$$I_L = \sqrt{3}\ I_{ph} = \sqrt{3} \times 25.98 = 45 \text{ A}$$

(iii) Active power

$$P = \sqrt{3}\ V_L I_L \cos\phi = \sqrt{3} \times 577.35 \times 45 \times 0.733 = 32.98 \text{ kW}$$

(iv) Reactive power

$$Q = \sqrt{3}\ V_L I_L \sin\phi = \sqrt{3} \times 577.35 \times 45 \times \sin(42.86°) = 30.61 \text{ kVAR}$$

A three-phase, 10 kVA load has a power factor of 0.342. The power is measured by the two-wattmeter method. Find the reading of each wattmeter when the (i) power factor is leading, and the (ii) power factor is lagging. **[May 2014]**

Solution

$$S = 10 \text{ kVA}$$

$$\text{pf} = 0.342$$

$$S = \sqrt{3}\, V_L I_L$$

$$10 \times 10^3 = \sqrt{3}\, V_L I_L$$

$$V_L I_L = 5.77 \text{ kVA}$$

$$\cos\phi = 0.342$$

$$\phi = 72°$$

(i) Reading of each wattmeter when the power factor is leading

$$W_1 = V_L I_L \cos(30° + \phi) = 5.77 \cos(30° + 70°) = -1 \text{ kW}$$

$$W_2 = V_L I_L \cos(30° - \phi) = 5.77 \cos(30° - 70°) = 4.42 \text{ kW}$$

(ii) Reading of each wattmeter when the power factor is lagging

$$W_1 = V_L I_L \cos(30° - \phi) = 4.42 \text{ kW}$$

$$W_2 = V_L I_L \cos(30° + \phi) = -1 \text{ kW}$$

Example 14

Two wattmeters are connected to measure power in a three-phase circuit. The reading of one of the wattmeters is 5 kW when the load power factor is unity. If the power factor of the load is changed to 0.707 lagging without changing the total input power, calculate the readings of the two wattmeters.

Solution (i) pf = 1, $W_1 = 5$ kW

(ii) pf = 0.707 (lagging)

(i) When the power factor is unity,

$$W_1 = W_2 = 5 \text{ kW}$$

$$\text{Power} = W_1 + W_2 = 5 + 5 = 10 \text{ kW} \quad (1)$$

(ii) When the power factor is changed to 0.707 lagging,

$$\text{pf} = \cos\phi = 0.707 \text{ (lagging)}$$

$$\phi = 45°$$

$$\tan\phi = \tan(45°) = 1$$

$$\tan\phi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2}$$

$$1 = \sqrt{3}\,\frac{W_1 - W_2}{10}$$

$$W_1 - W_2 = \frac{10}{\sqrt{3}} = 5.77 \text{ kW} \quad (2)$$

Solving Eq. (1) and (2),

$$W_1 = 7.89 \text{ kW}$$

$$W_2 = 2.11 \text{ kW}$$

$$I_{ph} = I_L = 8.03 \text{ A}$$

$$Z_{ph} = \frac{V_{ph}}{I_{ph}} = \frac{100}{8.03} = 12.45\,\Omega$$

Example 17

Each phase of a 3ϕ delta connected load has an impedance of $Z_{ph} = 50\angle 60°\Omega$. The line voltage is 400 V. Calculate the total power. What will be the reading of two wattmeters connected to measure the power. [Dec 2015]

Solution $Z_{ph} = 50\angle 60°\Omega$

$V_L = 400$ V

For a delta-connected load,

$$V_{ph} = V_L = 400 \text{ V}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{400}{50} = 8 \text{ A}$$

$$I_L = \sqrt{3}\, I_{ph} = \sqrt{3} \times 8 = 13.86 \text{ A}$$

(i) Total power

$$P = \sqrt{3}\, V_L I_L \cos\phi = \sqrt{3} \times 400 \times 13.86 \times \cos(60°) = 4.8 \text{ kW}$$

(ii) Reading of two wattmeters

$$W_1 = V_L I_L \cos(30° - \phi) = 400 \times 13.86 \times \cos(30° - 60°) = 4.8 \text{ kW}$$

$$W_2 = V_L I_L \cos(30° + \phi) = 400 \times 13.86 \times \cos(30° + 60°) = 0$$

Example 18

Three coils each with a resistance of 10 Ω and reactance of 10 Ω are connected in star across a three-phase, 50 Hz, 400 V supply. Calculate (i) line current, and (ii) readings on the two wattmeters connected to measure the power.

Solution $R = 10\,\Omega$

$X_L = 10\,\Omega$

$V_L = 400$ V

Example 17

Each phase of a 3ϕ delta connected load has an impedance of $Z_{ph} = 50\angle 60°\Omega$. The line voltage is 400 V. Calculate the total power. What will be the reading of two wattmeters connected to measure the power. **[Dec 2015]**

Solution

$$Z_{ph} = 50\angle 60°\Omega$$

$$V_L = 400 \text{ V}$$

For a delta-connected load,

$$V_{ph} = V_L = 400 \text{ V}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{400}{50} = 8 \text{ A}$$

$$I_L = \sqrt{3}\ I_{ph} = \sqrt{3} \times 8 = 13.86 \text{ A}$$

(i) Total power

$$P = \sqrt{3}\ V_L I_L \cos\phi = \sqrt{3} \times 400 \times 13.86 \times \cos(60°) = 4.8 \text{ kW}$$

(ii) Reading of two wattmeters

$$W_1 = V_L I_L \cos(30° - \phi) = 400 \times 13.86 \times \cos(30° - 60°) = 4.8 \text{ kW}$$

$$W_2 = V_L I_L \cos(30° + \phi) = 400 \times 13.86 \times \cos(30° + 60°) = 0$$

Example 18

Three coils each with a resistance of 10 Ω and reactance of 10 Ω are connected in star across a three-phase, 50 Hz, 400 V supply. Calculate (i) line current, and (ii) readings on the two wattmeters connected to measure the power.

Solution

$$R = 10\ \Omega$$

$$X_L = 10\ \Omega$$

$$V_L = 400 \text{ V}$$

For a star-connected load,

(i) Line current

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94\text{ V}$$

$$\bar{Z}_{ph} = R + jX_L = 10 + j10 = 14.14\angle 45°\ \Omega$$

$$Z_{ph} = 14.14\ \Omega$$

$$\phi = 45°$$

$$\text{Power factor} = \cos\phi = \cos(45°) = 0.707\text{ (lagging)}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{230.94}{14.14} = 16.33\text{ A}$$

$$I_L = I_{ph} = 16.33\text{ A}$$

(ii) Readings on the two wattmeters

$$P = \sqrt{3}\, V_L I_L \cos\phi = \sqrt{3} \times 400 \times 16.33 \times 0.707 = 7998.83\text{ W}$$

$$W_1 + W_2 = 7998.83\text{ W} \qquad (1)$$

Also,

$$\tan\phi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2}$$

$$\tan 45° = \sqrt{3}\,\frac{W_1 - W_2}{7998.83}$$

$$W_1 - W_2 = 4618.13\text{ W} \qquad (2)$$

Solving Eqs (1) and (2),

$$W_1 = 6308.48\text{ W}$$

$$W_2 = 1690.35\text{ W}$$